

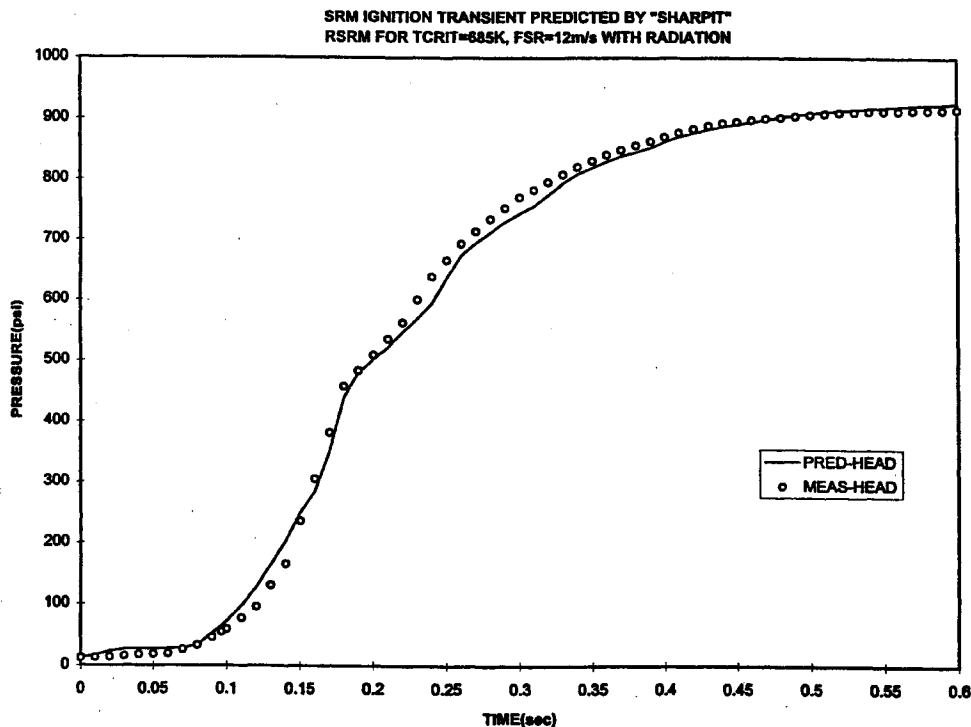


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Modern SRM Ignition Transient Modeling (Part 1): Introduction and Physical Models

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Modern SRM Ignition Transient Modeling **(Part 1): Introduction and Physical Models**

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Abstract

The rapid pressurization of a rocket chamber during its ignition transient leads to a rapidly changing flowfield and structural loading, and consequently has been the cause for many motor failures (e.g. Challenger, SRMU PQM1). Modeling of this phenomenon has for many years been conducted using volume-filling and one-dimensional analyses, and only recently two-dimensional analyses. Currently, three-dimensional fully-coupled flow/structural ignition models are being developed by a new generation of analysts. Unfortunately, a number of conference papers and journal articles have been published in recent years in which the predicted ignition transients did not match measurements. Often, the cause of this disagreement was blamed on numerical factors such as insufficient grid resolution when in reality the cause was inadequate modeling of the physics. Consequently, a series of papers has been written documenting the latest thinking on the modeling of ignition transients. This first paper in the series summarizes the history of ignition modeling and the equations of volume-filling and 1D methods, as well as what is known and not known about the physics of the ignition process; the use of experimental data from hoop strain gages and laser/furnace ignition tests is explained, and models are presented for wet and dry grain preheating, ignition criteria, propellant viscoelasticity, and pressurization of grain overhangs. Subsequent papers in the series¹⁻⁴ will present applications to a wide range of motors, and will describe the status of current multi-dimensional modeling efforts.

History of Modeling Efforts

The ignition of a solid rocket motor (SRM) causes a number of extreme events to occur in the combustion chamber within a fraction of a second: (1) pressure jumps to as high as 2000 psi, (2) temperature jumps about 3000°K, and (3) the propellant and case deform, especially with composite cases and soft propellants. The ignition also drives other critical phenomena: (1) case rotation and seal capability at intersegment joints, (2) ignition overpressure in silos and on launch pads, and (3) flyout during vehicle staging. Consequently, understanding and modeling the ignition transient process is crucial for developing reliable rocket motors.

Early ignition transient modeling was accomplished using volume-filling methods whereby the equation for conservation of mass for the rocket chamber was solved for the history of chamber pressure assuming isothermal filling. Propellant ignition was defined by specifying empirically-determined "ignition delay time" and "flame spreading

time". Since the chamber pressure was assumed spatially uniform, the analysis was termed "zero-dimensional".

Volume-filling methods were extended by Salita⁵⁻⁹ in a number of ways: (1) chamber temperature was determined in addition to pressure by solving the coupled mass and energy conservation equations, (2) the solution for chamber temperature allowed grain preheating to be calculated rather than imposed, (3) by subdividing the bore into N sub-volumes an approximate 1D model of flame-spreading was easily added, (4) a simple model for propellant ignition temperature was developed, (5) the effect of a film of water in delaying propellant ignition was modeled for steam-launched vehicles, (6) radiative grain preheating from the gas/droplet cloud in the chamber was modeled, and (7) igniter pressurization was coupled to that of the chamber. The simplicity of volume-filling methods allowed the estimation of case expansion effects which became important with the increasing use of composite case materials in the 1980s. Modeling of small L/D chambers of Space Motors is more appropriate with volume-filling than with a 1D model.

However, large boosters with large length-to-diameter (L/D) ratios generate large pressure drops down the bore that must be predicted using one-dimensional (1D) models. The first viable one-dimensional analysis was developed at Princeton by Peretz¹⁰. The unsteady equations of flow were solved on a computational domain subdivided into N axial bore segments. Convective preheating of the propellant grain from an axial pyrogen igniter was modeled, but radiation was ignored. Ignition of each propellant segment was predicted by estimating the history of surface temperature using Goodman's equation and by specifying a surface ignition temperature of 850°K. It will be shown below that this ignition temperature is much too high, and that radiative preheating is very important.

The Peretz analysis was extended by Lal/Santha/Padmanabhan¹¹ to account for canted-port igniters and by Pardue/Han¹² to account for two-phase flow. Rozanski¹³ specialized the modern 2D CFD code SHARP at Thiokol to generate 1D ignition simulator SHARP1DIT and used Salita's ignition criterion and wet grain modeling. Salita¹⁴ identified and quantified the propagation of the igniter shock down the bore of a Shuttle booster using hoop strain gage data.

As computing power improved in the early 1990s, 2D analyses began to appear in order to improve the modeling in the nozzle submergence and intersegment slot regions. Gehris/Price^{15,16} modified the Raven code at NWC, but the effort was later dropped. Johnston¹⁷ applied an inviscid

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model to Titan and Shuttle boosters, but was unable to match the Shuttle pressure trace. Ciucci/Jenkins/Foster¹⁸ modeled the flowfield created in the RSRM forward fin using a 2D flow model (the circumferential flow was assumed to be uniform in each slot), and showed good agreement with oil smears generated in cold-flow tests. A flame spreading predictive capability was added later¹⁹ but no comparison to data was made.

Huseman²⁰ applied the axisymmetric LMCO code IGNITE to model the ignition of RSRM, Titan SRMU, and SICBM motors. IGNITE is a fluid-in-cell finite-difference solution to the Euler (inviscid) unsteady equations, and allows for a two-gas formulation with moving boundaries (to allow for nozzle plug rupture). Both convective and radiative grain preheating were accounted for.

The interaction between the 2D flow and propellant deformation was studied both analytically and experimentally. Salita¹⁴ correlated and modeled the viscoelastic response of RSRM propellant during the ignition transient to its pressurization rate using hoop strain gage data. He also developed a simple flow analog to show that the pressure differential around a grain overhang at an intersegment slot would grow with the square-root of the overhang height; conversely, as chamber pressure rises during an ignition transient, the differential around the overhang grows, which increases flow stagnation ahead of the overhang, which increases the overhang, etc. This unstable coupling was the cause of the failure of the first static test of the Titan SRMU (PQM-1), and has been modeled by numerous analysts (see below).

The failure of PQM-1 led to a flurry of modeling activity. Chang/Patel/Yang²¹ were the first to couple an axisymmetric ignition transient flowfield solution to a propellant deformation calculation. Their viscous flow solution was time-marched through the ignition transient; however, the marching was stopped at nine times during the transient, the pressure field was passed to the ABAQUS code which calculated a new deformed grain shape, the flowfield was regridded, and the time marching restarted. With this analysis they were able to model the grain collapse in SRMU PQM-1. Johnston/Murdock^{22,23} automated this process so that the grain shape and flowfield grid were updated roughly every millisecond without stopping the flowfield solution. However, their model of propellant deformation was a correlation of quasi-steady predictions of ABAQUS; ignoring the time-dependent nature of the propellant viscoelasticity (dashpot) prevented them from matching accurately the grain collapse in PQM-1. Wang et al²⁴ extended the methodology of Johnston/Murdock by removing limitations on the description of the propellant surface and flowfield grid.

Limited 3D modeling has been attempted in the past. Rozanski/Loh (Thiokol) extended SHARPIT to three dimensions; unfortunately, funding ran out before the code was fully operational. Recently, however, the Center for Simulation of Advanced Rockets (CSAR) at the University

of Illinois has been modeling the fully-coupled structural/flow interaction in the RSRM during ignition and propellant burnback under a five-year contract with DOE. Alavilli et al²⁵ modeled the ignition transient of the Shuttle booster as part of this effort, but their predicted pressure rose too slowly due to specifying too high an ignition temperature and ignoring the importance of radiative grain preheating. These problems have been corrected in more-recent studies at CSAR²⁶.

Luke and Meyer² are simulating the ignition transient for Atlas 5 including the 3D flame spreading in the aft fins. Chan/Matheson/Jarymowczyk²⁷ of LMCO/Sunnyvale modeled D5 ignition by modifying the RAVEN code. All 3D models should be validated by simulating the aft-fin flame spreading measured by Kuo et al²⁸ for generic deep and shallow slot configurations.

Iverson/Graham³ are extending the original flaw propagation code SBAS-1 (originally called SBRAM and developed for SGI platforms) to SBAS-2 for Windows NT. SBAS couples both volume-filling and 1D bore ballistics to 3D propellant structural deformation and crack propagation codes to predict whether flaws (voids, debonds, cracks) would propagate fatally upon pressurization.

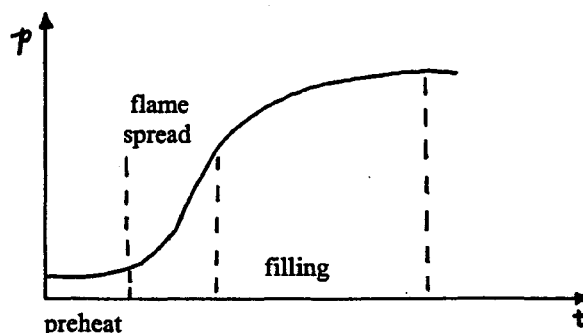


Fig 1 – Chamber pressurization during the ignition transient.

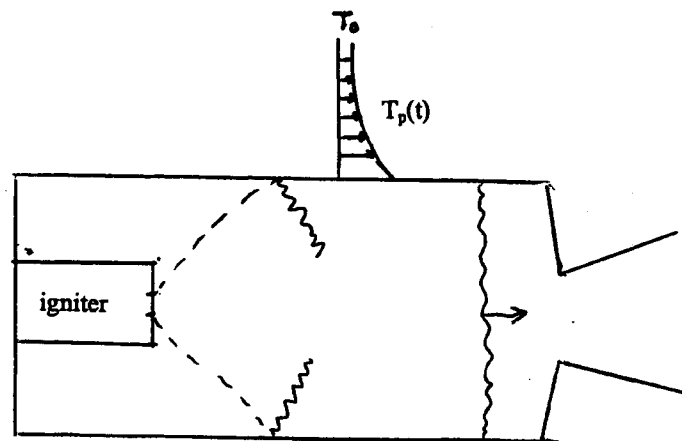


Fig 2 – Igniter shocks and grain preheating.

The Ignition Transient Phenomenon

Fig 1 shows a schematic of the pressure history at the head and aft ends of the motor during the ignition transient. After the igniter is fired, the main propellant grain rapidly preheats until surfaces near the igniter reach an ignition condition. The propagation of this ignition condition away from the igniter is called flame spreading. Flame spreading occurs radially outward in axial fins as well as axially along the bore. Generally, flame spreading is complete (the entire propellant surface has ignited) about halfway to full bore pressurization. Subsequently, the bore continues to pressurize until the mass flux from the propellant surface is just that value that can be passed through the choked nozzle. The resulting pressure at the nozzle throat will then be

$$p_c(t) = p_{ref} [(S/A^*) \rho_p r_{ref} C^*/p_{ref}]^{1/(1-n)} \quad (1)$$

where A^* is the eroding nozzle throat area, $S(b)$ is the burning surface area of the main propellant of density ρ_p whose burn rate is $r = r_{ref}(p/p_{ref})^n \epsilon$ where $\epsilon = \exp[(1-n)(T_o - T_{ref})/\tau_k]$ accounts for the effect of preconditioning temperature T_o , $b = \int r dt$ is the burnback distance, and $C^*(T_o)$ is the characteristic velocity of the propellant. In long motors, the mass addition along the bore causes the stagnation pressure p^o to drop, so that the head end pressure will be higher than that given by eq(1). This pressure drop can be estimated from 1D duct flow as a function of injected mass flux

$$dp^o/p^o = -\gamma M^2 dm_{in}/m_{in} \quad (2)$$

There are many types of igniters. Boosters usually use head end pyrogen igniters whose nozzles (ports) either face axially aftward or are canting to impinge directly onto the propellant surface. The impingement of the igniter plume on the propellant surface (which even occurs for most axial igniters) creates an oblique shock that enhances the heat flux to the propellant (Fig 2). Space motors may use aft end igniters, head end tangential igniters, or consumable wafer igniters⁴⁰. Tactical motors may use pyrogen igniters or bag igniters that spew hot particles onto the propellant surface to ignite it.

For pyrogen igniters, the rapid gas ejection from the igniter acts like a gaseous piston, and propagates compression waves down the bore that coalesce into a shock wave. When this shock reaches the aft closure, it partially reflects back toward the head end, and partially transmits down the nozzle after rupturing any weatherseal or plug in the nozzle. This process has been exposed by hoop strain gage data (Fig 3).

Lessons from Hoop Strain Gage Data

Bore pressure can be deduced at any axial location where a hoop strain gage is wrapped around the motor case during static firing¹⁴. Fig 3 shows the pressure histories deduced at five axial locations along the bore of the Shuttle QM-3 static test motor; it clearly reveals the propagation of the igniter shock down the port and its subsequent reflection from and rupture of the nozzle plug. Confidence in this data is

provided by the agreement in timing of the plug rupture with break-wire data, and with the speed of propagation predicted by 1D ignition codes.

Fig 4 presents a plot of hoop strain versus head end pressure at one location along the CP of the forward segment of the Shuttle booster. Salita¹⁴ showed that the looping behavior is caused by the viscoelastic behavior of the propellant, which could be modeled as a spring-dashpot with resiliency coefficient proportional to the bore pressurization rate. This says that propellant deformation caused by motor pressurization will only occur late in the ignition transient once dp/dt has diminished (as in a viscous damper). This explains why the grain collapse in SRMU PQM-1 occurred at the end of the ignition transient (Fig 5).

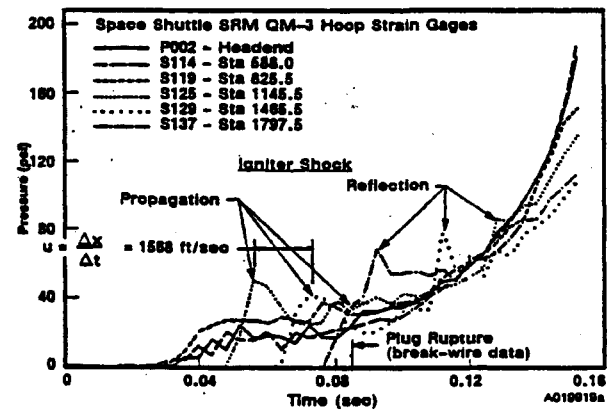


Fig 3 – Propagation of igniter shock deduced from hoop strain gage data (from Ref 14).

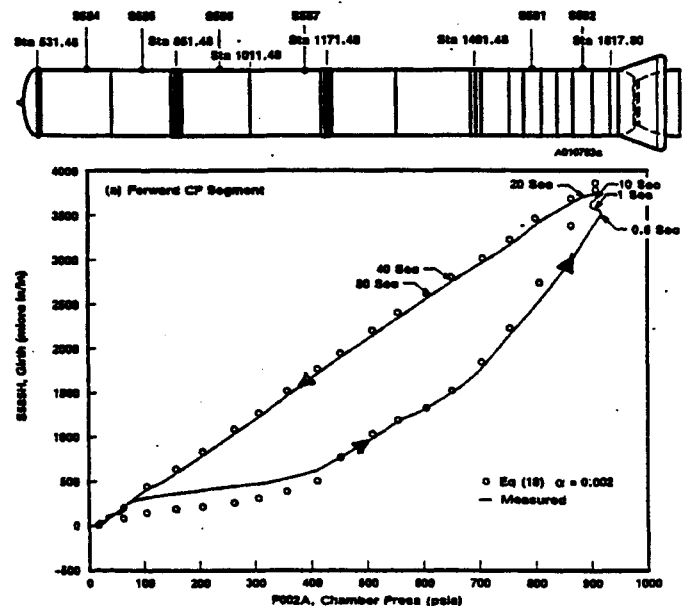


Fig 4 – Viscoelastic response of propellant during bore pressurization deduced from hoop strain gage data.

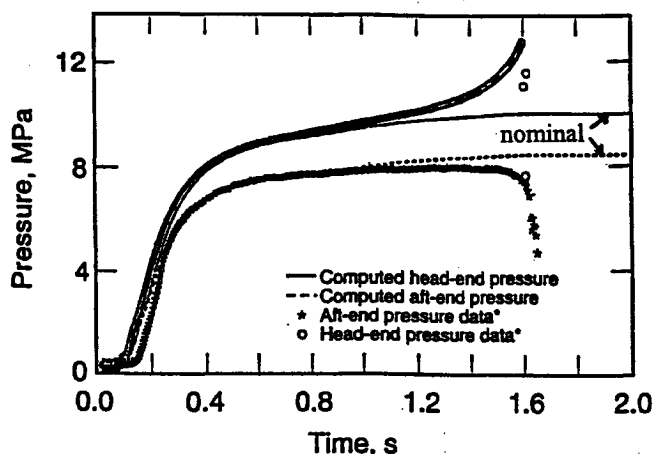


Fig 5 – Grain collapse in SRMU PQM-1 occurred after full bore pressurization (from Ref 22).

Ignition Transient Modeling Influences System Design

Ignition transient modeling has had a major impact on motor and system design and on program continuity. Three examples are presented below.

Modeling of the grain collapse in SRMU PQM-1 conducted at the Aerospace Corporation²¹⁻²⁴ was instrumental in identifying the fix, which involved rounding the forward face of the grain overhang at the intersegment slot.

The ability to model the ignition transient process had major consequences after the first Shuttle flight (STS-1). Shortly after the ignition of the solid boosters on STS-1, a “quasi-shock” wave reflected off the launch pad and nearly damaged the control surfaces on the orbiter. At first, NASA blamed this wave on the solid-booster igniter and requested that the igniter be redesigned to generate a softer ignition. However, using Fig 3 and other modeling tools, Salita showed that the igniter shock was long gone by the time the pressure wave passed over the orbiter; indeed, close examination of video of the launch showed the igniter shock passing through the debris cloud from the explosive rupture of the tie-down bolts at exactly the predicted time. The analysis further showed that the compression wave created by ignition of the main grain was the cause of the ignition overpressure (IOP) on the pad. Thus, ignition transient modeling prevented a costly and erroneous fix. Since redesign of the main grain was out of the question, suppression of the IOP was achieved externally by introducing a powerful water spray and water blanket system on the Mobile Launch Platform (MLP).

After ejection of a missile from a steam-launch cannister, the ignition delay was so long that fear of a hang-fire temporarily stopped the program. Salita added wet-grain effects to his ignition transient code VOLFIL to show that the ignition delay caused by the film of condensed water on the propellant surface could be quantified. This allowed the motor program to continue.

Physical Models

Flow Modeling

In short ($L/D \approx 1$) motors, the chamber pressure and temperature are fairly uniform, so that approximating them as time-dependent but spatially-independent (“zero-dimensional”) has usually been reasonable, especially when 1D flame spreading models were added. The resulting unsteady “lumped-parameter” model was easily solved and has served as the workhorse of the rocket community. In longer ($L/D \gg 1$) motors, a significant drop in stagnation pressure along the bore occurs due to mass addition from the propellant grain (eq 2); this drop is greatest when the bore radius is smallest, i.e. during the ignition transient. This pressure drop has led to the development of one-dimensional ignition-transient models, the first of which was developed by Peretz¹⁰. A second advantage of 1D models is their ability to track the ignition waves that are propagated down the bore and nozzle. The rocket community has recognized the value of both zero- and one-dimensional models.

However, for many years the rocket community chose not to extend ignition modeling to two and three dimensions. The disinterest in 2D modeling existed because the only two-dimensional flow occurs in the neighborhood of intersegment slots that exist only in a few large boosters; note that the flow in the cavity behind submerged nozzles is essentially 1D during the ignition transient because the submergence cavity is usually nearly full of propellant. Three-dimensional unsteady modeling, although useful for simulating the ignition process in fins, has until recently been avoided due to the extensive required computational resources that have been beyond those available to many organizations.

Salita’s lumped-parameter model VOLFIL4 and Rozanski’s 1D model SHARP1D-IT will now be summarized as examples of their genre; both codes were developed at Thiokol and are provided in the flow propagation code SBAS-1 available to the community³. Examples of 2D and 3D modeling are presented in Parts 2-5 of this series of papers.

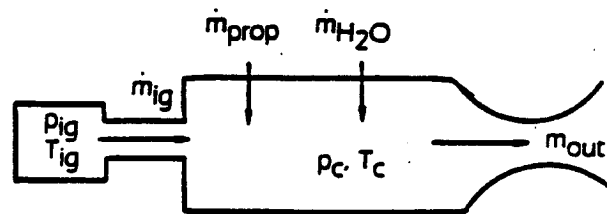


Fig 6 – Schematic of volume-filling analysis of motor ignition.

1) Volume-Filling Model VOLFIL4

Conservation of mass and energy for the rocket chamber of volume $V_c(t)$ provide two equations for the history of lumped (spatially-averaged) values of chamber pressure $p_c(t)$ and temperature $T_c(t)$ (Fig 6). For a single calorically-perfect gas:

$$dp_c/dt = [(RT_c/V_c)(m_{in}-m_{out}) + (p_c/T_c)dT_c/dt - (p_c/V_c)dV_c/dt]/(1+\beta p_c/V_c) \quad (3)$$

$$dT_c/dt = (RT_c/p_c C_v V_c)[(m_{in} C_{p, in} T_{p, in} - m_{out} C_p T_c) - (m_{in}-m_{out})(C_p-R)T_c] - Q_w/C_v \quad (4)$$

where the mass flow rates into and out of the chamber are

$$m_{in} = m_{ig} + \rho_p Sr$$

$$m_{out} = \alpha p_c A^*/C^*$$

and m_{ig} is the mass flow from the igniter, α is a factor that accounts for unchoked nozzle outflow, β is the rate of change of void volume with pressure due to case expansion, and Q_w is the rate of heat loss from the gas to the wall (propellant + case). If m_{ig} is not specified, it can be calculated from a mass conservation equation like eq(3):

$$dp_{ig}/dt = (RT_{ig}/V_{ig})[(\rho_p Sr)_{ig} - m_{ig}] \quad (5)$$

where the igniter volume V_{ig} and propellant flame temperature T_{ig} are assumed constant. For a large L/D motor that generates a drop in stagnation pressure along the bore, the lumped value p_c corresponds to the pressure at the nozzle throat since the outflow is determined by the nozzle throat area.

By subdividing the propellant surface into N axial segments, the surface temperature, time of ignition, and mass efflux on each segment are solved independently to provide a simple 1D model of flame spreading. In addition, the effect of a distribution of water film on the surface can be simulated. The volume-filling method has also proved effective for the tangential and consumable wafer igniters used in space motors.

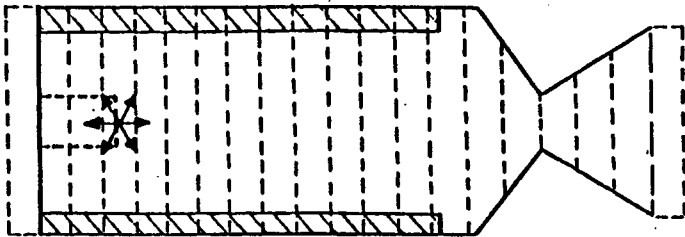


Fig 7 – Schematic of one-dimensional computational grid for motor ignition analysis.

2) One-Dimensional Analysis SHARP1D-IT

Many 1D ignition simulation codes have been written since the original Peretz work. In SHARP1D-IT, Rozanski¹³ formulated the unsteady one-dimensional flow equations in conservation form in order to capture the igniter shock:

$$\partial E/\partial t + \partial F/\partial X = W + \partial R/\partial X \quad (6)$$

$$\text{where } E = \begin{vmatrix} \rho \\ \rho U \\ e \end{vmatrix} A \quad F = \begin{vmatrix} \rho U \\ \rho U^2 + p \\ (e+p)U \end{vmatrix} A$$

where $e = p/(\gamma-1) + 1/2 \rho U^2$ and the source (injection) terms are

$$W = \begin{vmatrix} \rho_p Sr + m_{ig} \\ p dA - 1/2 \rho U^2 f A_w + m_{ig}^2 / \rho A \\ \rho_p Sr C_p T_r - Q_w A_w + m_{ig} C_{p, ig} T_{ig} \end{vmatrix} A$$

$$R = \begin{vmatrix} 0 \\ G \\ GU + k_g \partial T/\partial X \end{vmatrix} A \quad G = (4/3) \mu \partial U/\partial X$$

and f is a friction coefficient for the bore flow along wetted surface area A_w , and $p dA$ accounts for a change in port area. These equations are solved in SHARP1D-IT using a modern flux vector-splitting technique specialized from multidimensional CFD. SHARP1D-IT allows for the igniter gas to be injected anywhere along the motor bore, which is useful for igniter ports not located at the head end of the bore.

Grain Preheating

The propellant grain is rapidly heated to ignition by both the convective Q_c and radiative Q_r heat fluxes from the gas/droplet flow emitted by both the igniter and from upstream segments of main propellant that have already ignited. It is common to predict when the propellant will ignite by solving the 1D heat conduction equation and imposing an ignition criterion such as critical surface temperature.

Assuming constant thermal diffusivity α and conductivity k in the propellant, the variation of propellant temperature T_p with depth Y and time t is determined either numerically or analytically from

$$\partial T_p/\partial t = \alpha [\partial^2 T_p/\partial Y^2 - R] \quad (7)$$

where R accounts for radiative heating

$$R = \partial Q_r/\partial Y \quad Q = Q_c + Q_r$$

$$Q_o(t) = Q(Y=0, t) = Q_{co} + Q_{ro}$$

subject to the boundary conditions

$$T_p(Y, 0) = T_p(\infty, t) = T_{po}$$

$$(\partial T_p/\partial t)_{Y=0} = -Q_o(t)/k \quad \text{or} \quad Q_o(t) = h[T_g(t) - T_s(t)]$$

This form of the heat conduction equation allows for in-depth radiative heating that can occur for materials whose surfaces may be partially transparent, due for example to

water film or binder accumulation. The radiative heat flux that reaches the depth Y depends on its wavelength λ and the material's extinction coefficient $\beta(\lambda)$, and is defined by Beer's Law to be

$$q_r(Y, \lambda) = q_{ro}(\lambda) e^{-\beta(\lambda)Y} \quad (8)$$

Defining the spectral integrals

$$Q_{ro} = \int q_{ro}(\lambda) d\lambda \quad Q_r(Y) = \int q_r(Y, \lambda) d\lambda \quad (9)$$

then

$$R = -\int q_{ro}(\lambda) \beta(\lambda) e^{-\beta(\lambda)Y} d\lambda \quad (10)$$

In general, eq(7) must be solved numerically. However, if β is independent of wavelength so that $R = -\beta e^{-\beta Y} Q_{ro}$, and if the total incident heat flux Q_o is constant in time, then the history of surface temperature can be solved exactly from eq(7) as²⁹

$$T_s(t) = T_{po} + [Q_{co} + (1-W)Q_{ro}]F \quad (11)$$

where

$$\begin{aligned} F &= (4\alpha t/\pi)^{1/2}/k \\ W &= \pi^{1/2} [1 - \exp(\tau^2) \operatorname{erfc}(\tau)]/2\tau \\ \tau &= \beta(\alpha t)^{1/2} \end{aligned}$$

As expected, for an opaque propellant ($\beta=\infty$), $W=0$ and all radiative heating occurs at the surface; for a transparent propellant ($\beta=0$), $W=1$ so that all radiation passes through the propellant without altering its temperature. For water³⁰, $\beta(1/\text{cm}) \approx 10^{8\lambda(\mu\text{m})/3-4}$, which is opaque for $\lambda > 3\mu\text{m}$. When all heating is at the surface, eq(7) may be simplified approximately as

$$dT_s/dt = 4\alpha h^2 (T_g - T_s)^3 / [3(2T_g - T_s - T_o)(T_g - T_o)k^2]$$

For a piecewise linear variation $Q_o(t) = A_i + B_i t$ over successive time intervals Δt_i , an exact solution to eq(7) is²⁹

$$T_s(t) = T_o + (A_1 + 2B_1 t/3)F + (2F/3)\Sigma(B_{k+1} - B_k)(t - t_k)^{3/2}$$

and the summation on subscript k goes from 1 to $i-1$.

1) Radiative Heat Flux Q_{ro}

The form of $q_{ro}(\lambda)$ depends on its source. A CO_2 laser emits only in the narrow band around $\lambda=10\mu\text{m}$, while aluminum oxide droplets emit like black bodies whose spectral radiance lies primarily in the range $\lambda=0-6\mu\text{m}$:

$$(q_{ro})_{BB} = 8942/[e^{14388/\lambda T} - 1]\lambda^5 \text{ (cal}/\mu\text{m}\cdot\text{cm}^2\cdot\text{sec}) \quad (12)$$

for λ in μm and T in $^\circ\text{K}$. Integrating the black-body spectral radiance over wavelength yields the classical result for a gas/droplet cloud:

$$Q_{ro} = \epsilon\sigma(T_d^4 - T_s^4)$$

where $\sigma = 1.3549 \times 10^{-12} \text{ cal}/\text{cm}^2\cdot\text{sec}\cdot\text{K}^4$ is the Stefan-Boltzmann constant, and T_d is the temperature of the radiating Al_2O_3 droplets, which is usually chosen to equal the

propellant flame temperature T_f but could be significantly higher (see discussion below).

Salita⁹ has devised a simple model for the emissivity ϵ of a cylindrical gas/droplet cloud in a rocket chamber assuming single isotropic scattering. This emissivity is strongly dependent on the size distribution of the Al_2O_3 droplets in the cloud, which Salita³¹ has shown for a wide range of aluminized propellants over a wide range of combustion pressures to be bimodal log-normal with (1) the small-diameter mode ("smoke") comprising most of the oxide mass (e.g. 80% for the Space Shuttle Booster propellant) and always a mass median diameter $D_m = 1.5\mu\text{m}$ with standard deviation $\sigma \approx 0.4$, and (2) the large-diameter mode ("coarse") comprising the remainder of the oxide mass having a mass-median diameter $D_m = 50-150\mu\text{m}$ that varies with pressure and propellant, but always with a standard deviation around $\sigma \approx 0.2$. Although the emissivity would appear to depend on numerous parameters, Salita showed that the smoke dominates the emissive power of the gas/droplet cloud due to its combination of high mass fraction and small diameter; this allows the emissivity to be written in terms of only two parameters, the droplet temperature T_d and a single concentration parameter $\kappa = 1.5(R_w/D_m)(\rho_{\text{mix}}/\rho_d)\Phi f_s$ that combines the effect of bore radius R_w , mass-median diameter D_m , cloud density ρ_{mix} , droplet bulk density ρ_d , Al_2O_3 flow fraction Φ , and smoke mass fraction f_s . The resulting Mie calculations were curve-fit approximately by the quadratic

$$\epsilon = 0.79[1 - 0.215(4.05T_D - T_D^2)]\kappa/(1 + 0.84\kappa) \quad (13)$$

where $T_D = T_d(\text{K})/1000$. The density of Al_2O_3 was approximated³² by $\rho_d(\text{gm/cc}) = 5.632 - 0.001127T_d(\text{K})$ ($= 1.8 \text{ gm/cc}$ at 3400°K). For RSRM whose CP sections have approximately a 30-inch radius and $\Phi = 0.288$ and $f_s = 0.80$, and assuming that the smoke is at the flame temperature T_f of the propellant, then $\kappa \approx 680(p/1000)$ and $\epsilon = 0.4-0.5$ during the ignition transient:

$p(\text{psi})$	$\rho_{\text{mix}}(\text{gm/cc})$	$T_f(\text{K})$	$\rho_d(\text{gm/cc})$	κ	ϵ
100	0.00071	3222	2.000	62	0.39
300	0.00210	3328	1.881	196	0.45
500	0.00347	3373	1.830	333	0.47
700	0.00484	3402	1.800	472	0.49
900	0.00620	3422	1.775	613	0.50

With $\epsilon \approx 0.5$ and noting that $(T_f/T_d)^4 \ll 1$, then

$T_d(^\circ\text{K})$	$Q_r(\text{cal}/\text{cm}^2\cdot\text{sec})$
3000	55
3500	102
4000	173
4500	278

Thus, for $T_d = T_f = 3500^\circ\text{K}$, $Q_r = 100 \text{ cal}/\text{cm}^2\cdot\text{sec}$, which will dominate the convective heating early in the ignition transient.

However, recent studies by Melcher et al³³ suggest that the temperature of the radiating gas/particle cloud can be much hotter than the propellant flame temperature; indeed, the flame zone around the burning Al droplet, which is the source of the Al_2O_3 smoke, could be as hot as 4400°K. This would drive the droplet density down and Q up, so that ϵ would approach unity. In addition T_d would increase by a factor of 2.5, so that the radiative heat flux could be as much as five times greater than for the assumption $T_d=T_f$.

2) Convective Heat Flux Q_{co}

Earlier models of convective heat transfer utilized modified forms of the Dittus-Boelter model for pipes. Newer models account for enhanced heating due to impingement of the igniter plume on the propellant surface³⁴:

$$h_{con} = [C_1 + C_2 F(X)] Re^{0.8} (A_p/A^*)^{0.4} (k/D_p) G \quad (14)$$

where $A_p = \pi D_p^2/4$ is the local port cross-sectional area.

Propellant Ignition Criterion

Different types of ignition criteria have been proposed and have been summarized by Kuo⁷; they include critical surface temperature, rate of increase of critical surface temperature, critical surface heat flux, and critical in-depth temperature. Critical surface temperature is the simplest and is usually used. A value of 850°K was originally chosen by Peretz, but propellant ignition data indicates that much lower values (550-750°K) are appropriate, as shown below.

Ignitability data is often generated for a propellant by heating a sample at constant flux either in a xenon arc-image furnace or with a CO_2 laser. After a short time of heating, surface reactions begin, as evidenced by "first light". However, if the heat source is then removed the reactions would stop. If heating is continued, eventually a "go/no-go" condition is reached beyond which the reactions continue even if the heat source is removed. Go/no-go times are shown in Fig 8 for a typical propellant, and are seen to depend on pressure as well as heat flux Q_o . For higher pressures (greater than about 250 psi), the time to go/no-go ignition may be curve-fit in the form

$$(t_{ig})^{1/2} = B/Q_o^c \quad (15)$$

The surface temperature at ignition can then be determined analytically²⁹. Eq(11) shows that the history of temperature $T_s(t)$ at the surface of an opaque propellant initially at temperature T_o but being heated at constant total flux Q_o is given by

$$T_s(t) = T_o + [4\alpha t/\pi]^{1/2} (Q_o/k) \quad (16)$$

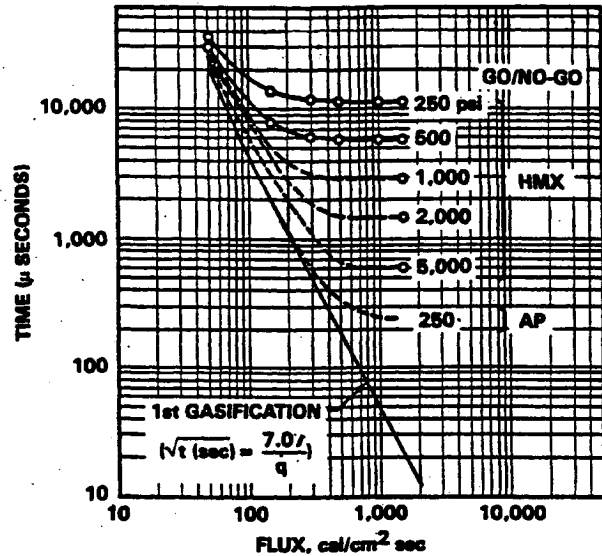


Fig 8 – Ignition behavior of solid propellant as a function of pressure and heat flux.

where k is the thermal conductivity and $\alpha = k/\rho_p C_p$ the thermal diffusivity of the propellant. Typical values for solid propellants are $\rho_p = 110 \text{ lbm/ft}^3 = 1.762 \text{ gm/cc}$ and $C_p = 0.29 \text{ BTU/lbm-R} = 0.29 \text{ cal/gm-K}$
 $k = 6.11 \times 10^{-5} \text{ BTU/ft-sec-R} = 9.09 \times 10^{-4} \text{ cal/cm-sec-K}$
 $\alpha = 1.915 \times 10^{-6} \text{ ft}^2/\text{sec} = 0.00178 \text{ cm}^2/\text{sec}$.

Combining eqs(15,16) predicts the surface temperature at ignition $T_{ig} = T_s(t_{ig})$ to be

$$T_{ig} = T_o + [4\alpha/\pi]^{1/2} (B/k) Q_o^{1-c} \quad (17)$$

For example, Baer and Ryan^{36,37} determined that for a wide range of AP-composite propellants and pressures, $B=5.5$ and $C=0.92$ for Q_o in $\text{cal/cm}^2\text{-sec}$ and t in sec. Thus, the Baer/Ryan data suggests that the ignition temperature T_{ig} depends only weakly on heat flux ($\sim Q_o^{0.08}$), so that eq(17) can be used even for variable flux. For a typical propellant initially at $T_o=300^\circ\text{K}$, propellant ignition temperatures are significantly lower than the 850°K assumed by many analysts:

$Q_o(\text{cal/cm}^2\text{-sec})$	$T_{ig}(^\circ\text{K}) = 300 + 288 Q_o^{0.08}$
10	646
100	716
200	740
500	773

However, at lower pressures (early in the ignition transient), eq(15) is inaccurate, and attempts have been made to quantify the surface reaction process in terms of competing finite rates (e.g. Price/Boggs³⁸).

For a given propellant, the value of B can vary significantly due to the surface condition (Fig 9) or heat source (Fig 10).

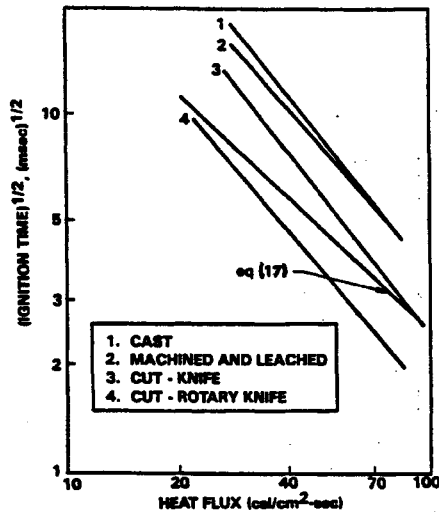


Fig 9 – Effect of propellant surface condition on ignition.

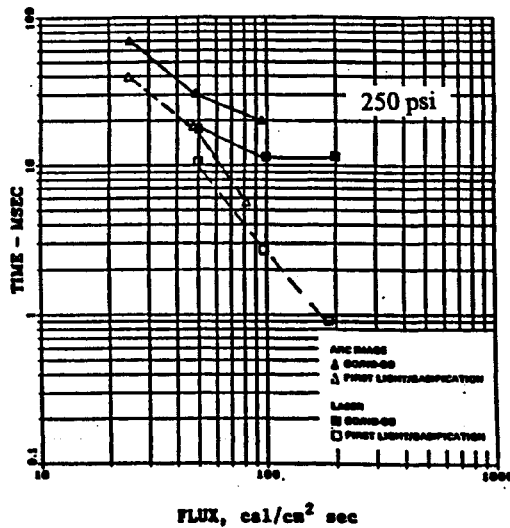


Fig 10 – Effect of heat source on propellant ignition times.

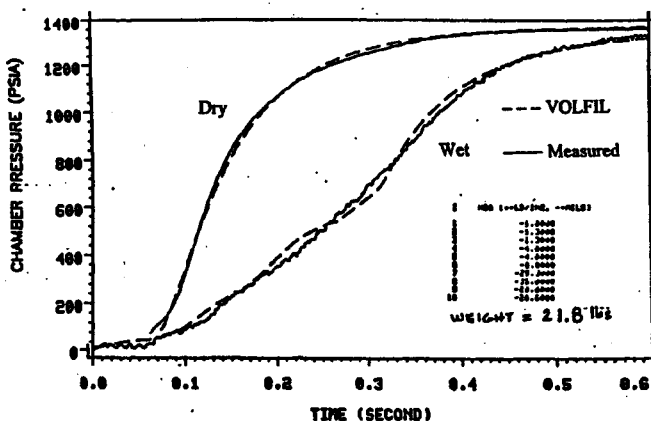


Fig 11 – Comparison of ignition transient predicted by VOLFIL4 for steam launch to that measured in full-scale static test.

Effect of Water Film on Propellant Surface

Sometimes a film of water will lie on the propellant surface during motor ignition. For example, during steam launch, steam that is injected into the motor bore condenses on the propellant to form a film of water of thickness $h(x)$. Some of this water might be absorbed immediately by the propellant, and the rest will be removed by the igniter flow due both to convective shearing and to vaporization. A model was developed by Salita⁸ to predict the delay in propellant ignition caused by the water film. It assumed that the ignition delay was too short to allow any absorption of water into the propellant, that the water film was too thin to allow removal by convective shear, and that the water film was opaque enough to radiation that all radiative flux would be absorbed by the water and none could pass through to preheat the grain until all of the water had been vaporized (although eq(10) with $\beta=10^{8\lambda(\mu\text{m})/3-4}$ predicts that about 75% of the incident energy flux penetrates a typical film thickness of 20 mils at 3500°K). The resulting three-phase model is summarized as follows:

Phase 1: The total heat flux Q raises the temperature of the water film according to $dT_{H_2O}/dt = Q/(\rho h C_p)_{H_2O}$ until its saturation temperature T_{sat} is reached.

Phase 2: The total heat flux then vaporizes the film at a rate $d(\rho_{H_2O}h)/dt = (Q - Q_s)/h_{fg}$ where h_{fg} is the enthalpy of water evaporation and $Q_s = (\rho h C_p)_{H_2O} dT_{sat}/dt$ is the heat flux required to maintain the water at its saturation temperature as chamber pressure rises.

Phase 3: The total heat flux only now begins to preheat the propellant after the water film has been completely vaporized.

Steam-table data³⁵ can be curve-fit with pressure according to

$$T_{sat} (^{\circ}\text{R}) = 101.74 + 50.514(\log_{10}p) + 33.711(\log_{10}p)^2$$

$$h_{fg} (\text{BTU/lbm}) = 36.3 - 109.854(\log_{10}p) + 80.227(\log_{10}p)^2 - 32.573(\log_{10}p)^3$$

The model described above was applied to a static test of a full-scale motor whose propellant surface was to be sprayed with water moments before ignition. The initial distribution of water-film thickness $h(x)$ had been previously measured in a cold-water injection test in an inert full-scale model of the motor bore. The resulting pressure transient predicted by VOLFIL4 agreed closely with the measured transient (Fig 11).

Effect of Grain Overhang

When the propellant segments of multi-segment motors are stacked, a gap (radial slot) is intentionally created. The bore radius at the forward end of a segment is purposely designed to be less than that at the aft end of the previous segment in order to allow for mandrel removal during casting of the segment and progressive rather than instantaneous burnout of a cylindrical bore. This differential in bore radius across the intersegment slot creates a grain "overhang". If the stagnation pressure on the forward face of the overhang is too high and the propellant modulus is too small, the grain can collapse into the bore as happened in the first static firing of the Titan IV SRMU booster²¹. The quasi-steady axisymmetric flowfield in the vicinity of the aft slot in RSRM has been calculated using a variety of CFD codes^{14,39} and is shown in Fig 12. Salita¹⁴ showed that the pressure differential predicted by the CFD codes around the overhang corner could be estimated using a correlation of measurements taken on forward facing steps:

$$p - p_{\infty} = C(74h/L)^{1/2} (R_L/1.5 \times 10^{-6})^{0.1} (1/2 \rho_{\infty} M_{\infty}^2) \quad (18)$$

where

$$C = \begin{matrix} 0.40 & \text{on forward face of overhang} \\ -0.75 & \text{on bore surface just aft of slot} \end{matrix}$$

where h is the overhang length, L is the length of the segment upstream of the overhang, R_L is the Reynolds number, and the subscript ' ∞ ' implies a centerline value just upstream of the intersegment slot. As the overhang increases, the pressure differential around the corner increases and the propellant grain begins to deform aftward and inward toward the bore centerline, which increases the overhang. If the grain modulus is low enough, the grain will eventually rupture and plug the bore as happened in the first test of the Titan SRMU (PQM-1)²¹⁻²⁴.

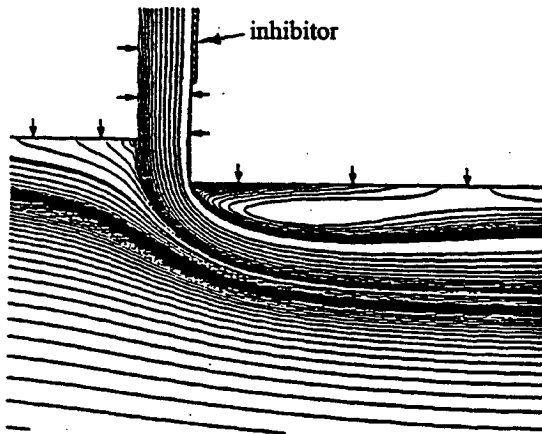


Fig 12 – Flow past a grain overhang at an intersegment slot.

Effect of Case Expansion and Joint Rotation

The motor void volume and propellant surface areas increase upon pressurization, especially for composite cases. A 70% increase in void volume during the ignition transient is not uncommon for composite cases. Indeed, even for RSRM with steel case, the width of the intersegment slots and their overhangs increase by a factor of 2-4 during pressurization¹⁴.

The effect of joint rotation had to be addressed in the 1980s because of the evolution of large multi-segment boosters (Titan, RSRM, Ariane). For segmented cases whose joints are much thicker than their membranes, the bulge will be greater between the joints (along the membranes) than at the joints, much like a balloon whose circumference is confined at midlength. This causes "joint rotation" even in steel cases like RSRM. Indeed, this was the cause of the Challenger accident, because the radial gap at the O-ring groove increased faster than the cold O-ring could follow.

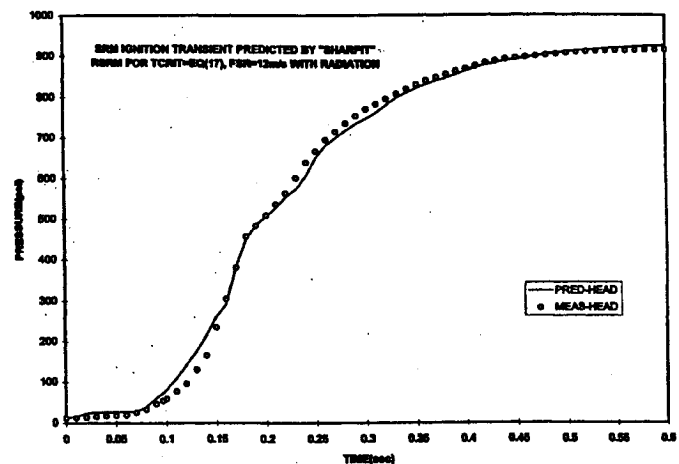


Fig 13 – Agreement between the RSRM measured pressure history and prediction of SHARPIDIT using eq(17) and Baer/Ryan data as an ignition criterion with radiation.

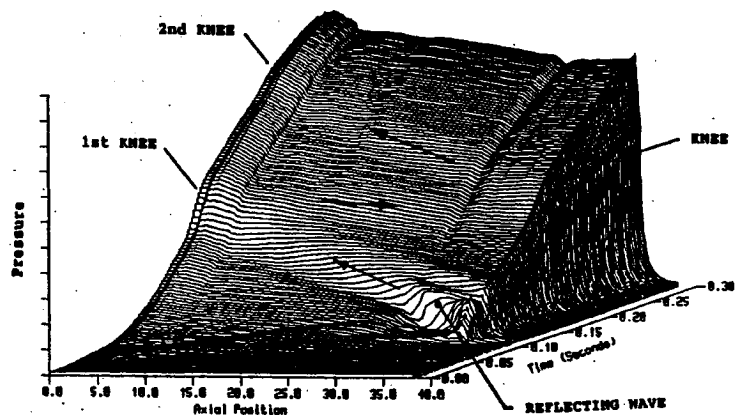


Fig 14 – Waterfall plot showing the reflections of the igniter shock up and down the RSRM bore.

Sample Case

The Space Shuttle Booster (RSRM) provides an interesting test of ignition modeling due in part to the double inflection points ("knees") in the measured pressure trace.

Fig 13 shows a comparison of the measured head end pressure history to that predicted by SHARP1DIT for the Shuttle Booster using the models of convective and radiative preheating described above, with eq(17) to model propellant ignition temperature and ignition constants B and C taken from the Baer/Ryan data. The only free constant was the flame-spreading rate into the forward fins, which was chosen as 12 m/sec. Agreement was excellent, especially in predicting the knees in the pressure trace at about 180 and 230 ms.

These knees are caused by the arrival of the first and second reflections of the igniter shock from the chamber aft end. The first reflection at the aft end was seen in Fig 3, but both reflections from the head end are seen clearly in a waterfall plot (Fig 14) of the SHARP1DIT predictions. The slightly early ignition between 100 and 150 ms is due to the single-gas limitation in SHARP1DIT. The properties of the main propellant (TP-H1148) were used, which included 16% aluminum loading. This is quite appropriate once a small portion of the main grain has ignited. However, until that time the igniter gas/droplet mixture with its lower aluminum content (10%) will generate a lower radiative flux.

Eq(17) for ignition temperature predicted that propellant ignition occurred at a surface temperature of 650-715°K, depending on bore location and heat flux. A single constant value of 685°K generated essentially the same ignition transient. Note that this value is significantly lower than the value of 850°K used by some analysts.

Fig 15 shows that ignoring radiative grain preheating causes a significant delay in the predicted propellant ignition.

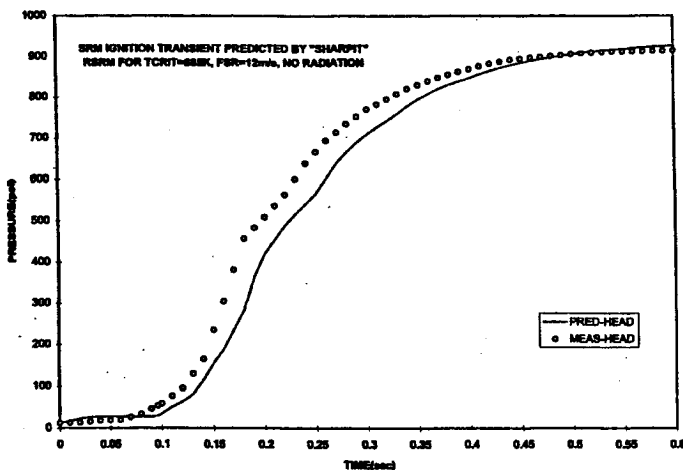


Fig 15 - Disagreement between the RSRM measured pressure history and prediction of SHARP1DIT using eq(17) and Baer/Ryan data but without radiative preheating.

Recommendations

Modelers of ignition transients should follow the recommendations of the other papers in this series¹⁻⁴ and should heed the following guidelines:

- 1) If the igniter shock does not appear in the numerical solution, there is something wrong.
- 2) Radiative grain preheating is important. Accounting for it by simply enhancing the convective heating ignores its geometric and time dependencies.
- 3) The propellant ignition temperature T_{ig} for constant heat flux Q_0 is identified by eq(17) where B and C should be determined experimentally. For AP-composite propellants at higher pressures, (a) $C \approx 1$ so that T_{ig} is nearly independent of Q_0 , and (b) $B \approx 5.5$ for Q_0 in $\text{cal/cm}^2\text{-sec}$. At lower pressures, eqs(15,17) must be modified.
- 4) Structural/ballistic interaction is not a quasi-steady phenomenon, but must account for the viscoelastic (time dependent) behavior of the propellant.

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